

$$\inf_{\hat{\theta}} \sup_{\theta \in \Theta} \mathbb{E} \{ \Phi(\rho(\hat{\theta}, \theta)) \} \geq \Phi(\delta) \cdot \inf_{\Psi} \mathbb{P}(\Psi(Z) \neq J)$$

$\hat{\theta}$ $\theta \in \Theta$ $\Phi(\delta)$ Ψ $\mathbb{P}(\Psi(Z) \neq J)$

2 δ -sep. set $\{\theta^{(1)}, \dots, \theta^{(M)}\}$

$$\forall i \neq j \quad \rho(\theta^{(i)}, \theta^{(j)}) \geq 2\delta$$

$$J \sim U\{1, \dots, M\}$$

$$Z|J \sim P_{\theta^{(J)}}(\cdot)$$

$$\text{prob. of error} \leftarrow q_e = \mathbb{P}(\Psi(Z) \neq J)$$

hypothesis testing

Fano Inequality

$$\forall \Psi: \quad q_e = \mathbb{P}(\Psi(Z) \neq J) \geq 1 - \frac{\log 2 + I(Z; J)}{\log M}$$

$$H(X) = H(P_X) := - \sum_{x_i \in \mathcal{X}} P_X(x_i) \log P_X(x_i)$$

$$H(X, Y) := - \sum_{x_i, y_j} P(x_i, y_j) \log P(x_i, y_j)$$

$$H(X|Y) := - \sum_{y_j} P(y_j) \underbrace{\sum_{x_i \in \mathcal{X}} P(x_i|y_j) \log P(x_i|y_j)}_{\text{Entropy of } P(\cdot|y_j)}$$

Weighted Sum

$$I(X; Y) := H(X) - H(X|Y) \leftarrow$$

$$= H(Y) - H(Y|X)$$

$$= D_{KL}(P_{X,Y} \parallel P_X \cdot P_Y) \geq 0$$

$$= \sum_{x,y} P_{X,Y} \log \frac{P_{X,Y}}{P_X \cdot P_Y}$$

$$= \underbrace{\sum_{x,y} P_{X,Y} \log \frac{P_{X,Y}}{P_Y}}_{P(Y) \cdot P(X|Y)} - \underbrace{\sum_{x,y} P_{X,Y} \log P_X}_{\sum_x P_X \log P_X}$$

$$= -H(X|Y) - H(X)$$

$$D_{KL}(P_{X,Y} \parallel P_X \cdot P_Y) = I(X; Y) \geq 0$$

$$= H(X) - H(X|Y) \geq 0$$

$$\Rightarrow H(X) \geq H(X|Y)$$

Conditioning would reduce the entropy

chain rule

$$P_{X,Y} = P_{X|Y} P_Y$$

$$H(X, Y) = H(X|Y) + H(Y) = H(Y|X) + H(X)$$

$$= - \sum_y P(y) \sum_x P(x|y) \log P(x|y) - \sum_y P(y) \log P(y)$$

$$= - \sum_{x,y} p(x,y) \log p(x|y) - \sum_{x,y} p(x,y) \log p(y)$$

$$= - \sum_{x,y} p(x,y) \log \underbrace{\{p(x|y) \cdot p(y)\}}_{p(x,y)} = H(X,Y)$$

$$H(X,Y|Z) = H(X|Y,Z) + H(Y|Z)$$

Fano's Ineq. Proof:

$$V := \mathbb{I}(Y(Z) \neq J) \quad Z, J$$

$$H(Z,J|V) = H(Z|J,V) + H(J|V)$$

$$= H(J|Z,V) + H(Z|V)$$

$$\underbrace{H(J,V|Z)}_A = \underbrace{H(V|Z,J)}_{0 \text{ b/c } V \text{ is a deterministic func. of } Z \& J} + H(J|Z)$$

$$= \underbrace{H(J|V,Z)}_{\text{no-error}} + H(V|Z)$$

$$\begin{aligned} & P(V=0) \underbrace{H(J|V=0,Z)}_{\leq \log(M-1)} \\ & + \underbrace{P(V=1)}_{q_e} \underbrace{H(J|V=1,Z)}_{\leq \log(M-1)} \end{aligned}$$

$$\begin{aligned} A = H(J|Z) &\leq q_e \log(M-1) + H(V|Z) \\ &\leq q_e \log(M-1) + H(V) \end{aligned}$$

$$b/c \quad H(V|Z) \leq H(V) = H(q_e) := -q_e \log q_e - (1-q_e) \log(1-q_e)$$

$$\underbrace{H(J)}_{\log M} - \underbrace{H(J)}_{-I(Z;J)} + H(J|Z) \leq q_e \log(M-1) + H(q_e)$$

$$\leq q_e \log M + \underbrace{H(1/2)}_{\log 2}$$

$$q_e \geq \frac{\log M - I(Z;J) - \log 2}{\log M} = 1 - \frac{I(Z;J) + \log 2}{\log M} \quad \square$$

$$\inf_{\hat{\theta}} \sup_{\theta \in \Theta} E\{\Phi(p(\hat{\theta}, \theta))\}$$

$$\textcircled{m} \geq \Phi(\delta) \cdot \left[1 - \frac{I(Z;J) + \log 2}{\log M} \right]$$

$$\text{Lemma:} \quad I(Z;J) \leq \frac{1}{M^2} \sum_{j,l=1}^M D_{KL}(P_{\theta^{(j)}} \| P_{\theta^{(l)}})$$

Proof:

$$I(Z;J) = D_{KL}(P_{Z,J} \| P_Z \cdot P_J)$$

$$= \sum_{z,j} P_{z,j} \log \frac{P_{z,j}}{P_Z \cdot P_J}$$

$$= \sum_{z,j} P_{z|J} \cdot \underbrace{P_J}_{\cancel{P_J}} \log \frac{P_{z|J} \cdot \cancel{P_J}}{P_Z \cdot \cancel{P_J}}$$

$$= \frac{1}{M} \sum_{j=1}^M \sum_z P_{\theta^{(j)}}(z) \log \frac{P_{\theta^{(j)}}(z)}{\frac{1}{M} \sum_j P_{\theta^{(j)}}(\cdot)}$$

$$= \frac{1}{M} \sum_{j=1}^M D_{KL} (P_{\theta^{(j)}} \| \underbrace{(\bar{P})}_{\substack{\text{Marginal dist. of} \\ Z}})$$

Recall that for a convex f

$$f(\alpha x + (1-\alpha)y) \leq \alpha f(x) + (1-\alpha)f(y)$$

$D_{KL}(P \| q)$ is convex w.r.t. q

$$= \sum_{x_i} P_{(x_i)} \log \frac{P_{(x_i)}}{q(x_i)} \quad \frac{\partial D_{KL}}{\partial q(x_i)} = P(x_i) \frac{-\frac{P(x_i)}{q^2(x_i)}}{\frac{P(x_i)}{q(x_i)}} = -\frac{P(x_i)}{q(x_i)}$$

$$\frac{\partial^2 D_{KL}}{\partial q^2(x_i)} = \frac{P(x_i)}{q^2(x_i)} > 0$$

$$\bar{P} \text{ is convex combination} = \frac{1}{M} \sum_{j=1}^M P_{\theta^{(j)}}$$

$$I(Z, J) = \frac{1}{M} \sum_{j=1}^M D_{KL}(P_{\theta^{(j)}} \| \bar{P})$$

$$\leq \frac{1}{M} \sum_{j=1}^M \underbrace{\frac{1}{M} \sum_{l=1}^M}_{\text{convex combination}} D_{KL}(P_{\theta^{(j)}} \| P_{\theta^{(l)}})$$

$$= \frac{1}{M^2} \sum_{j, l=1}^M D_{KL}(P_{\theta^{(j)}} \| P_{\theta^{(l)}})$$

□

Example : $\mathcal{N}(\theta, \sigma^2)$ $\{0, 2\delta, -2\delta\}$ be the 2δ -sep set.

$$m \geq \delta^2 \left[1 - \frac{I(z; J) + \log 2}{\log 3} \right]$$

$$D_{KL}(\mathcal{N}(\theta^{(i)}, \sigma^2) \parallel \mathcal{N}(\theta^{(j)}, \sigma^2))$$

$$= \frac{n(\theta^{(i)} - \theta^{(j)})^2}{2\sigma^2} \rightarrow \frac{n 4\delta^2}{2\sigma^2} = \frac{2n\delta^2}{\sigma^2}$$

$$\text{or } \frac{8n\delta^2}{\sigma^2}$$

$$\text{or } 0$$

$$I(z; J) \leq \left[0 + 0 + 0 + \frac{8n\delta^2}{\sigma^2} \times 2 + \frac{2n\delta^2}{\sigma^2} \times 4 \right] / 9$$

$$= \left(\frac{8}{3} \right) \frac{n\delta^2}{\sigma^2}$$

$$m \geq \delta^2 \left[1 - \frac{\frac{8}{3} \frac{n\delta^2}{\sigma^2} + \log 2}{\log 3} \right]$$

